Khadijah Aamir Chaudhry

Computer Science · Secure Systems · Applied Research
Lahore, Pakistan · +92 345 4430787 · khadijah.aamir.ch@gmail.com
LinkedIn · GitHub · Portfolio

EDUCATION

Lahore University of Management Sciences (LUMS) 2023–2027
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School 2018–2023
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS Aug 2025–present
Supervisor: Dr. Muhammad Ali Siddiqi

- To study runtime hardware Trojan detection, built a simulated Rocket Chip/Verilator testbed that captures RISC-V transaction traces.
- To assess post-quantum methods on constrained devices, benchmarked three algorithms on ESP32 for latency, memory, network cost, and estimated energy.

Research Assistant, SysNet Lab, LUMS Jan–May 2026
Supervisor: Dr. Naveed Anwar Bhatti

- To explore accessible eyesight screening, co-developed EyeSense's smartphone ToF and machine-learning pipeline.
- Supported lab and eye-camp data collection. Across 120 participants, the team reported 1.15 D mean absolute error.

SELECTED PROJECTS

On Chip Behavioral Monitoring · write-up 2026–present

- To prepare data for runtime monitoring research, set up four RV32 configurations and captured transaction traces with Verilator.
- Replayed benign and trigger-active firmware, creating trace data for later detector evaluation.

EyeSense · write-up Jan–May 2026

- To investigate phone-based eyesight checks, co-developed a ToF phase-map and machine-learning pipeline for refractive-error screening.
- The team evaluated the approach with 120 participants and reported 1.15 D mean absolute error, connecting sensing research to a human screening need.

Post-Quantum Cryptography on ESP32 · code Aug–Dec 2025

- To assess cryptography on low-resource hardware, benchmarked Kyber-512, Light-SABER, and Dilithium-II on ESP32.
- Compared latency, memory, network cost, and estimated energy to clarify trade-offs for embedded use.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS Aug 2026–present

- To support about 450 first-year students, designed and conducted programming labs and prepared and graded quizzes.
- Held weekly office hours and invigilated exams, providing recurring help during the course.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions Jul 2026–present

- To give the consultancy a consistent public identity, coordinated team delivery of its brand kit and office collateral.
- Coordinated LinkedIn launch content and the website so prospective clients can understand services and get in touch.

Data Collection Coach, Uplift AI Jul–Aug 2024

- To support a Pakistani-language AI dataset, led 10 data collectors and organized field outreach.
- Ran daily data-quality reviews and feedback sessions to improve consistency in the collected material.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS Aug 2026–present

- To help 15 incoming students adjust to university, led orientation and faculty introductions.
- Continued monthly first-year support sessions to offer students an ongoing point of contact.

Women in STEM Coordinator, LUMS SPADES Sep 2024–May 2025

- To widen participation in STEM, helped launch SPADES' first women-only STEM event at PSIFI.
- More than 100 participants attended the event and took part in a dedicated STEM program.

Co-Founder and Logistics Director, Umeed-e-Nau NGO Sep 2021–May 2023

- To create inclusive learning opportunities, co-founded an NGO serving underprivileged and disabled children.
- Organized educational activities and visits that enabled children to learn and participate together.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

100% Merit Scholarship, Lahore Grammar School · Aug 2021–May 2023
Shortlisted writing submission, LUMSx Nexus · May 2025
Honorable Mention, EducationUSA Pakistan Writing Contest · Sep 2022

TECHNICAL SKILLS

Programming: C · C++ · Python · JavaScript · Java · Assembly
Embedded and systems: ESP32 · Tiva C · MIPS · RISC-V · Rocket Chip · Verilator · FreeRTOS
Web and mobile: React · Android · Firebase Authentication · Firestore
Methods and tools: ML/AI · CNNs · PCA · K-means · ToF sensing · Docker · Git · post-quantum benchmarking